Sachin Thakur

sachint.2306@gmail.com sachin-thakur-22558017b https://thakur-sachin.github.io/sachin_portfolio/

Professional Experience

Robotics Engineer, ML, *Luminous Robotics* Jan 2026 – Present | South Boston, MA, USA

- Built and led the sim-to-real **digital twin** of an autonomous solar-panel installation robot in **NVIDIA Isaac Sim 6**: installer and pallet-carrier robots running the real **ROS 2** stack with simulated perception, behavior trees, Nav2-style navigation (MPPI) and **nvblox** mapping in the loop.
- Kept the twin faithful to production: sim-only behavior isolated in its own config, a written sim/real difference list, measured real-time factor (~0.15×) for timeouts, and audits proving merges leave production unchanged.
- Developed GPU-accelerated arm motion planning with **cuRobo/cuMotion**, including a direct pick-to-place panel carry with a torque-aware wrist strategy that cut peak J5 gravity load **0.96 → 0.80**; validated in sim, then on a real **Stäubli** arm.
- Added collision-aware "curtain" checks on live sensor maps, a 2D obstacle grid for navigation and a nav safety monitor.
- Shipped features across **~10 repositories** (messages, motion planning, behavior trees, config, deploy images, React/TypeScript operator UI), each opt-in behind one UI switch; ran multi-repo merge waves and released sim builds v1.1–v1.2.1.
- Built automated sim regression runs on GPU cloud (**AWS EC2, Docker, NICE DCV**) with live parameter read-back and placement scoring; **halved sim boot time** by persisting the shader cache and scoping the S3 asset sync.
- Field tests on production robots: outdoor 3D reconstruction with GPS; joint-torque study of legacy planner vs. cuMotion.
- Built, as one of two engineers, the data pipeline that helped win the **AWS and NVIDIA Physical AI grant** (1 of 9 globally).
- Fine-tuned pose estimators (**FoundationPose, EfficientPose, CenterPose, DOPE**) for robust real-world performance.

Machine Learning Scientist, *Oxipital AI* Jun 2024 – Dec 2025 | Bedford, MA, USA

- Implemented ReMaX-inspired panoptic segmentation optimizations (ReMask + ReClass) in **PyTorch**, cutting training time 4→3 days, maintaining Panoptic Quality (PQ), and reducing false positives by 15%.
- Developed a **DINOv3-style SSL backbone** for panoptic segmentation; built data infra for training a foundation vision model.
- Redesigned the vision transformer for memory-efficient, variable-resolution training, enabling higher-resolution inputs that increased **PQ by 5%** at comparable training cost.
- Designed **binary and sparse HDF5 representations** for high-volume sensor data, reducing storage by **~60%**.
- Led redesign of the inference pipeline across **GPU, container, and on-prem environments**: **300ms → 200ms** with FP16 and ONNX/TensorRT, then **~110ms** after **NVIDIA Nsight**-guided optimization.

Education

M.S. Mechanical Engineering, *Columbia University* — Minor: Robotics and Control Aug 2023 – Dec 2024 | New York, USA

- Courses: Intro to Robotics, Artificial Intelligence, Applied Robotics, Robot Studio, Control Theory, Evolutionary Algorithms

B.E. Mechanical Engineering, *Birla Institute of Technology & Science* Aug 2018 – May 2022 | Rajasthan, India

Projects

Trex – Adaptive Bipedal Robot, *Technologies: Isaac Sim, Isaac Lab, MuJoCo, PyTorch, PPO, Raspberry Pi*

- Designed and 3D-printed a 1.6 kg, **10-DoF biped** in Fusion 360; trained walking policies with **asymmetric PPO in Isaac Sim 6** and a bit-identical **Isaac Lab** port, using only onboard IMU and servo feedback.
- Fitted a servo model to the robot's walking logs (**1.4° RMS**) and checked every policy in **MuJoCo** through the exact Raspberry Pi control loop; sim-trained policies now **walk on the real robot**.

Autonomous Quadruped Project

- Built an autonomous quadruped fusing **IMU, LIDAR, and cameras** for navigation; secured a **$15,000 BITSAA grant**.

Smart Traffic Monitoring System, *Technologies: YOLOv5, Flask, IoT, RasPi, OpenCV*

- Designed an **IoT data pipeline** from **Raspberry Pi edge devices** into an on-prem Flask service, with buffering and visualization built for **intermittent connectivity and constrained compute**.

Collaborative Delivery Robots, *Technologies: OpenCV, SLAM, Path Planning, Micro-controllers*

- Developed a delivery-robot swarm with **multi-agent path planning** and **mono-camera SLAM** for a national competition.

Publications

End-to-End Deep Reinforcement Learning Platform for Competitive Physical Dynamic Systems, *NeurIPS* Dec 2024

- Built a high-fidelity Isaac Sim foosball simulator with realistic dynamics and contact modeling to train AI agents.

CPPS for monitoring, visualization, and control, *IJCIM* Mar 2023

- Designed a **tiered edge–fog–cloud data architecture** for real-time sensor monitoring and severity-based storage.

Skills

- **Deep Learning & CV:** PyTorch, TensorFlow, OpenCV, ViTs, ConvNeXt, YOLO
- **Optimization:** ONNX, TensorRT, FP16/INT8, pruning, Nsight, DDP
- **Languages:** Python, C++, Bash, TypeScript/React, FastAPI
- **Robotics & Sim:** ROS 2, Isaac Sim, cuRobo/cuMotion, Nav2, nvblox, MuJoCo
- **Control & Planning:** RL (PPO, A3C), MPC, MPPI, PID, behavior trees
- **Infra & Data:** Docker, Kubernetes, AWS (EC2, S3), MLflow, HDF5, MCAP